

An Explicit 26-Measurement Phase-Retrieving Frame in $\mathbb{C}^8$

Proof Candidate

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Abstract

Can 26 squared absolute values of complex linear functionals distinguish every vector in $\mathbb{C}^8$ up to a common phase? We give an explicit Gaussian-integer construction and a self-contained candidate proof that they can. A vector is represented by a polynomial in a specified eight-dimensional space. Three missing coefficients make 26 positive samples determine its squared modulus. To show that the polynomial is then determined up to phase, we compare two possible solutions through their average and difference. Equal moduli impose pointwise real orthogonality. Successive coefficients reduce this condition to positive sums of squares and, in the last case, a real symmetric two-by-two matrix whose determinant is strictly negative. The claimed conclusion is $m_{\mathbb{C}}(8) \leq 26$, with no assertion of optimality. The identities and frame entries have passed local review and separate exact symbolic checks; external independent expert review is pending.

Status. Proof candidate for examination; not peer reviewed. The complete proposed proof and exact input are given below and in the supplementary reproducibility archive.

1 The problem and an explicit construction

A measurement of a vector $x \in \mathbb{C}^8$ will mean a squared modulus $|q^*x|^2$, where $q^*x = \sum_k \overline{q_k}x_k$. Every such measurement is unchanged when x is multiplied by a complex number of modulus one. The question is whether 26 choices of q can distinguish every pair of vectors except for this unavoidable common phase. A family with this property is called a phase-retrieving frame. We construct one explicitly.

We identify a vector x with a polynomial f_x in a particular eight-dimensional space. The measurement vectors will lie on a single polynomially parametrized curve: their measurements are simply four times the values $|f_x(t)|^2$ at positive real parameters t . The construction has to achieve two different things. It must make 26 samples determine the entire polynomial $|f_x(t)|^2$, and it must make that polynomial determine f_x up to common phase.

Here is the argument in advance. The squared moduli belong to a 26-dimensional real polynomial space, because three coefficients always vanish. Rolle's theorem then makes any 26 distinct positive samples sufficient. To recover the polynomial itself from its squared modulus, we align the first nonzero coefficients of two possible solutions and compare their average v and half-difference h . Equal squared moduli say that $v(t)$ and $h(t)$ are perpendicular in the real plane $\mathbb{C}$ for every real t . Their first two coefficients leave only four cases. In three

cases a positive sum of squares forces the difference to vanish. The last case reduces to a real symmetric two-by-two matrix whose diagonal entries have opposite signs; its determinant can never be zero. The calculation below supplies the exact entries of this matrix and covers all cases, including vanishing initial coordinates.

For a polynomial $f(t) = \sum c_k t^k$, write $f^\#(t) = \sum \overline{c_k} t^k$. Thus $ff^\#$ is a real polynomial and equals $|f(t)|^2$ when t is real. We call it the squared-modulus polynomial of f .

The basis below is assembled from three short polynomials and a shifted copy of them. Write $a = (a_0, a_1, a_2)^T$ and set

$$a = \begin{pmatrix} 1 + it + t^3 + 4it^5 \\ t^2 + \frac{i}{2}t^3 - t^5 \\ t^4 + it^5 \end{pmatrix}, \quad J = \begin{pmatrix} 8 & 0 & 2 \\ 0 & 1 & 0 \\ 2 & 0 & 0 \end{pmatrix}.$$

The matrix J records a coefficient of the symmetric product: $[t^5](aa^T) = iJ$, where $[t^5]$ denotes the coefficient of t^5 . This identity explains the choice of J ; Section 2 shows how it produces the required cancellation.

Take the three entries of $a + t^8 J^{-1} a^\#$, followed by $t^6 - 2it^7$, the three entries of $t^8 a$, and finally t^{14} . The conjugated correction is chosen to preserve the three missing coefficients of the squared modulus. It also prevents the low-degree part and its shifted copy from producing a simple ambiguity.

Indeed, without that correction the comparison space would contain both a_0 and $t^8 a_0$. The two nonproportional polynomials

$$a_0(1 + it^8), \quad a_0(1 - it^8)$$

would then have identical squared moduli on the real line. The correction breaks this particular construction of an ambiguity; the proof below must still exclude all other pairs. The coefficients, including the middle coefficient $-2i$, are an explicit choice for which the subsequent strict inequalities hold. Expanded, the eight basis elements are

$$\begin{aligned} p_0(t) &= 1 + it + t^3 + 4it^5 + \frac{1}{2}t^{12} - \frac{i}{2}t^{13}, \\ p_1(t) &= t^2 + \frac{i}{2}t^3 - t^5 + t^{10} - \frac{i}{2}t^{11} - t^{13}, \\ p_2(t) &= t^4 + it^5 + \frac{1}{2}t^8 - \frac{i}{2}t^9 + \frac{1}{2}t^{11} - 2t^{12}, \\ p_3(t) &= t^6 - 2it^7, \\ p_4(t) &= t^8 + it^9 + t^{11} + 4it^{13}, \\ p_5(t) &= t^{10} + \frac{i}{2}t^{11} - t^{13}, \\ p_6(t) &= t^{12} + it^{13}, \\ p_7(t) &= t^{14}. \end{aligned} \tag{1}$$

Let $t_1, \dots, t_{26}$ be any distinct positive real numbers, and set

$$q_j = 2(\overline{p_0(t_j)}, \dots, \overline{p_7(t_j)})^T \in \mathbb{C}^8. \tag{2}$$

Theorem. For all $x, y \in \mathbb{C}^8$,

$$|q_j^* x|^2 = |q_j^* y|^2 \quad (1 \leq j \leq 26) \quad \implies \quad y = e^{i\theta} x$$

for some real θ . In particular, take $t_j = j$; then all entries of the 8-by-26 matrix $Q = (q_1, \dots, q_{26})$ are Gaussian integers.

Every q_j is nonzero, since its last entry is $2t_j^{14}$. Writing $m_{\mathbb{C}}(8)$ for the smallest possible number of such measurements, the theorem gives $m_{\mathbb{C}}(8) \leq 26$. It does not assert that 25 measurements are impossible. Each measurement is given by the single vector q_j in (2), or equivalently by the rank-one matrix $q_j q_j^*$. No more general quadratic measurements are used.

2 Why 26 positive samples determine the squared modulus

Put $U = \text{span}_{\mathbb{C}}(p_0, \dots, p_7)$ and $f_x = \sum_{k=0}^7 x_k p_k$. The orders of vanishing of these basis polynomials at zero are $0, 2, \dots, 14$, and their first coefficients are all 1. They are therefore linearly independent: in any nonzero linear combination, the term of lowest order cannot cancel. In particular $x \mapsto f_x$ is an identification of $\mathbb{C}^8$ with U .

A squared-modulus polynomial has degree at most 28 and would normally have 29 real coefficients. The three linear restrictions below explain the number 26 in this construction. Here $[t^k]$ means the coefficient of t^k .

For every $u \in U$,

$$[t^1](uu^\#) = [t^5](uu^\#) = [t^{13}](uu^\#) = 0. \quad (3)$$

These identities hold for all complex coefficients of u . They can be checked directly from (1), or by verifying that the three coefficients vanish in all products $p_j p_k^\#$. The exact checker does the latter, checking real and imaginary parts separately. For example, writing $u = \sum z_j p_j$, its first six coefficients are

$$z_0, iz_0, z_1, z_0 + \frac{i}{2}z_1, z_2, 4iz_0 - z_1 + iz_2.$$

Consequently the coefficient of t^5 in its squared-modulus polynomial is

$$2 \operatorname{Re}(4i|z_0|^2 - z_1 \bar{z}_0 + z_0 \bar{z}_1 + \frac{i}{2}|z_1|^2) = 0.$$

To verify the degree-13 cancellation, return to the short vector a and the real symmetric matrix J used to assemble (1). The corrected low-degree entries are $a + t^8 J^{-1} a^\#$, the shifted entries are $t^8 a$, and the remaining two are $t^6 - 2it^7$ and t^{14} . Direct multiplication of the three short polynomials gives

$$[t^5](a_i a_j^\#) = 0, \quad [t^5](a_i a_j) = iJ_{ij}.$$

In products of two corrected low-degree entries, the coefficient of degree 13 comes from the two cross terms. Their coefficient matrices are iJJ^{-1} and $-iJ^{-1}J$, whose sum is zero. A product of a first-group entry and a conjugated tail entry has coefficient $[t^5](a_i a_j^\#) = 0$. Products involving the middle entry $t^6 - 2it^7$ have no degree-13 term: the low portions have degree at most 12, the high portions start at degree 14, and the middle entry's own norm is $t^{12} + 4t^{14}$. The remaining products start beyond degree 13. This proves the three cancellations in (3) directly.

Lemma 1 (positive sampling). A nonzero real polynomial with at most m nonzero monomials has at most $m - 1$ distinct positive zeros.

Proof. Induct on m . Divide by its lowest power of t , which does not change positive zeros. The resulting polynomial has a nonzero constant term and at most $m - 1$ other monomials.

Its derivative has at most $m - 1$ monomials and therefore at most $m - 2$ distinct positive zeros by induction. Rolle's theorem proves the claim. The case $m = 1$ is immediate. $\square$

The space of real polynomials of degree at most 28 with the three coefficients in (3) equal to zero has dimension 26. The point of Lemma 1 is that evaluation at any 26 distinct positive numbers is injective on this space; dimension counting alone would not establish that fact.

If the measurements of x, y agree, set $f = \sum x_j p_j$ and $g = \sum y_j p_j$. By (2), $q_j^* x = 2f(t_j)$ and $q_j^* y = 2g(t_j)$, so the measurements are four times the corresponding squared moduli. Their difference of squared-modulus polynomials $ff^\# - gg^\#$ has degree at most 28 and misses the three exponents in (3). It therefore has at most 26 nonzero monomials. It vanishes at the 26 distinct positive sample points, so Lemma 1 gives

$$ff^\# = gg^\#. \quad (4)$$

It remains to prove that (4) has only proportional solutions in U .

3 Comparing two polynomials with the same squared modulus

We now prove that (4) determines a polynomial in U up to common phase. The order of vanishing at zero organizes the comparison. The nested spaces $\text{span}(p_i, \dots, p_7)$ consist of the elements of U vanishing to order at least $2i$. Every nonzero element belongs to exactly one of the differences between successive spaces.

If either polynomial in (4) is zero, both are zero. Otherwise let i be the first nonzero basis index of f . Its squared-modulus polynomial starts in degree $4i$, with coefficient equal to the squared modulus of that basis coefficient. The same must be true of g . Hence both have the same first index and leading coefficients of equal modulus. Multiply one by a unit complex number to align these coefficients, and divide both by the resulting common nonzero coefficient. The squared-modulus identity is preserved, and the coefficients at index i are now both 1.

Set

$$v = \frac{f + g}{2}, \quad h = \frac{f - g}{2}.$$

At every real t , the exact identity $|v(t) + h(t)|^2 - |v(t) - h(t)|^2 = 4 \text{Re}(h(t)\overline{v(t)})$ shows what equal measurements require: the average and the difference must be perpendicular in the real plane $\mathbb{C}$ at every parameter. This is an equality for the original pair of polynomials, not a linear approximation to it. Since their first coefficients have been aligned, h vanishes to higher order than v . With all parameters below real, we can write the following. Here x_j, y_j are new real coordinates of the normalized average v , not the coordinates of the two original signals; r_j, a_j are the real and imaginary coordinates of h . These a_j are scalar coordinates, unrelated to the short polynomials used to assemble the basis in Section 1.

$$\begin{aligned} v &= p_i + \sum_{j>i} (x_j + iy_j) p_j, \\ h &= \sum_{j>i} (r_j + ia_j) p_j, \\ F(t) &:= hv^\# + h^\#v = 0. \end{aligned} \quad (5)$$

The index i ranges over all eight possibilities. In particular, we have not assumed that the first coordinate of the original vector is nonzero. Nor have we divided by a polynomial or by a

measurement value. Common factors, repeated zeros, and zero measurement values remain within the argument.

The next coefficient distinguishes most of the remaining possibilities. After its initial power of t is removed, each p_j starts at 1 and has a purely imaginary first derivative. Write this derivative as is_j . Thus

$$p_j = t^{2j}(1 + is_j t) + O(t^{2j+2})$$

where the real numbers s_j are

$$(s_0, \dots, s_7) = (1, \frac{1}{2}, 1, -2, 1, \frac{1}{2}, 1, 0). \quad (6)$$

Lemma 2 (comparison of the first two coefficients). If $h \neq 0$ in (5), and j is its first nonzero index, then $r_j = 0$ and $s_j = s_i$.

Proof. The first two potentially nonzero coefficients of F are $2r_j$ and, after $r_j = 0$, $2(s_i - s_j)a_j$. No larger basis index can contribute to these two degrees. If the slopes differ, both parts of the supposedly first nonzero coefficient vanish, a contradiction. $\square$

The lemma says that the first possible nonzero difference has to have the same first imaginary derivative as the leading basis element of the average. From (6) the complete list is:

first signal index i	possible first difference indices
0	2, 4, 6
1	5
2	4, 6
3	none
4	6
5, 6, 7	none

For $i = 3, 5, 6, 7$, no nonzero h is possible. It remains to consider $i = 0, 1, 2, 4$. The next section gives every equation needed for these four cases.

4 The coefficient equations force the difference to vanish

For each first index i , put

$$W_i = \text{span}_{\mathbb{C}}(p_{i+1}, \dots, p_7), \quad W_7 = \{0\}.$$

After normalization we have $v \in p_i + W_i$ and $h \in W_i$. For each fixed v , consider the real-linear map

$$L_v : W_i \longrightarrow \mathbb{R}[t]_{\leq 28}, \quad L_v(h) = hv^{\#} + h^{\#}v,$$

where the complex vector space W_i is regarded as a real vector space. Our remaining task is to prove $\ker L_v = \{0\}$ for every such v . Pointwise perpendicularity alone would not imply this: for example, $h = icv$ with $c \in \mathbb{R}$ satisfies $hv^{\#} + h^{\#}v = 0$. The normalization excludes this nonzero constant-phase direction because h has no p_i coefficient whereas v has coefficient 1. The rest of the argument uses the requirement that h belongs to the fixed polynomial space W_i .

The low-order coefficients of $L_v(h)$ determine most coordinates of h successively; a few higher-order coefficients rule out the remaining coordinates. Thus all calculations below concern the kernel of one linear map at a time, for an arbitrary normalized average v .

Write $E_k = [t^k]F$. To extract any of these equations directly, write $v(t) = \sum_{\ell} v_{\ell} t^{\ell}$ and $h(t) = \sum_{\ell} h_{\ell} t^{\ell}$; these are ordinary monomial coefficients, with coefficients outside the degree range understood to be zero. Then

$$E_k = 2 \operatorname{Re} \sum_{\ell=0}^k h_{\ell} \overline{v_{k-\ell}}.$$

After the preceding variables have been determined, the equation $E_{2i+2j} = 0$ has coefficient 2 on r_j and determines it uniquely. The next equation has coefficient $2(s_i - s_j)$ on a_j . If this number is nonzero, it determines a_j as well. If it is zero, we retain a_j and keep any resulting equation as a necessary condition. These operations apply to every actual solution of $F = 0$.

Every division is by a fixed nonzero rational number, never by a variable expression that might vanish for a special signal. Thus no exceptional cases are lost. All identities below follow by polynomial multiplication and these substitutions. The exact checker reconstructs them from (1), but the required equations are displayed here.

The order of the four cases puts the shortest calculations first. Most end with a nonzero difference coefficient times a strictly positive sum of squares. In the last case two coefficients remain together; their two equations form a symmetric matrix with a determinant of fixed sign.

4.1 The average starts with p_4

By Lemma 2, a nonzero difference can first occur only at index 6. In this case write

$$\begin{aligned} v &= p_4 + (x_5 + iy_5)p_5 + (x_6 + iy_6)p_6 + (x_7 + iy_7)p_7, \\ h &= (r_6 + ia_6)p_6 + (r_7 + ia_7)p_7. \end{aligned}$$

Here is the short elimination in full. The first equation is $E_{20} = 2r_6 = 0$. With $r_6 = 0$, the coefficient E_{21} vanishes identically because the relevant slopes agree, and the next two are

$$E_{22} = 2r_7 + 2a_6y_5, \quad E_{23} = 2a_7 - a_6x_5.$$

Setting these equal to zero gives

$$r_6 = 0, \quad r_7 = -a_6y_5, \quad a_7 = \frac{1}{2}a_6x_5.$$

After these substitutions,

$$E_{25} = \frac{a_6}{2} (x_5^2 + 2(y_5 - 2)^2 + 8). \quad (7)$$

The factor in parentheses is strictly positive. Thus $a_6 = 0$, and all coefficients of h vanish.

4.2 The average starts with p_1

Only differences starting at 5 need consideration. Eliminating the coefficients of degrees 12 through 17 gives

$$r_5 = 0, \quad r_6 = -a_5 y_2, \quad a_6 = a_5 x_2, \quad r_7 = -a_5 y_3, \quad a_7 = 5a_5 x_3.$$

The following coefficient is then a particularly useful necessary constraint:

$$E_{21} = -2a_5 \left(1 + x_2^2 + y_2^2 + 10x_3^2 + 2y_3^2 \right). \quad (8)$$

It forces $a_5 = 0$, hence $h = 0$. Unused coefficients are not presumed to vanish automatically: it is sufficient that every actual solution must satisfy (8).

4.3 The average starts with p_2

Start with differences at 4 or later. Processing degrees 12 through 16 gives

$$\begin{aligned} r_4 &= 0, \\ r_5 &= -a_4 y_3, \\ a_5 &= 6a_4 x_3, \\ r_6 &= a_4 \left(-1 - y_4 + \frac{5}{2}y_3 - 5x_3 y_3 \right). \end{aligned}$$

The degree-17 equation is

$$E_{17} = -a_4 \left(30x_3^2 + 5(y_3 - \frac{2}{5})^2 + \frac{46}{5} \right). \quad (9)$$

Thus $a_4 = 0$, and the coefficients of h at 4 and 5 are zero. Restarting with a possible difference at 6, degrees 16 through 19 give

$$r_6 = 0, \quad r_7 = -a_6 y_3, \quad a_7 = 3a_6 x_3.$$

Now

$$E_{21} = -2a_6(1 + 6x_3^2 + 2y_3^2). \quad (10)$$

Hence $a_6 = 0$, and again $h = 0$.

4.4 The average starts with p_0 : the first possible difference

Process degrees 2 through 8. The solved variables are

$$\begin{aligned} r_1 &= a_1 = r_2 = 0, \\ r_3 &= -a_2 y_1, \quad a_3 = \frac{1}{6}a_2 x_1, \\ r_4 &= a_2 \left(1 - y_2 - \frac{5}{2}y_1 + \frac{5}{6}x_1 y_1 \right). \end{aligned}$$

The degree-9 coefficient becomes

$$E_9 = a_2 \left(\frac{5}{6}x_1^2 + 5(y_1 - \frac{2}{5})^2 + \frac{46}{5} \right). \quad (11)$$

It follows that $a_2 = 0$. Consequently every coefficient of h before index 4 is zero.

4.5 The average starts with p_0 : the last two coefficients

Now start h at index 4. The coefficients of degrees 8 through 15 give

$$\begin{aligned} r_4 &= 0, & r_5 &= -a_4 y_1, & a_5 &= a_4 x_1, \\ r_6 &= -a_4 y_2, \\ r_7 &= -a_4 y_3 - a_6 y_1 + a_4 x_2 y_1, \\ a_7 &= 3a_4 x_3 + \frac{1}{2}a_6 x_1 - \frac{1}{2}a_4 x_1 x_2. \end{aligned} \tag{12}$$

There are two remaining independent imaginary difference variables, a_4, a_6 . Introduce the real polynomials

$$\begin{aligned} R &= x_1^2 + 2(y_1 - 2)^2 + 8, \\ T &= -2x_1 x_3 - 4y_1 y_3 - 2x_2, \\ S &= -1 - x_1^2 - y_1^2 - 6x_3^2 - 2y_3^2 - x_2^2 - y_2^2, \\ \rho &= a_6 - a_4 x_2. \end{aligned} \tag{13}$$

After (12), direct coefficient extraction gives

$$E_{17} = \frac{R\rho + 2a_4 T}{2}, \quad E_{21} = 2a_4 S + T\rho. \tag{14}$$

The last two equations have a simple two-dimensional meaning. In the coordinates $(2a_4, \rho)$ for the remaining difference, (14) reads

$$\begin{pmatrix} S & T \\ T & R \end{pmatrix} \begin{pmatrix} 2a_4 \\ \rho \end{pmatrix} = \begin{pmatrix} E_{21} \\ 2E_{17} \end{pmatrix}.$$

The two diagonal entries have fixed opposite signs: $S \leq -1$ and $R \geq 8$. The associated quadratic form is therefore negative on one coordinate axis and positive on the other. In this two-dimensional system the off-diagonal entry T , whatever its sign, contributes $-T^2$ to the determinant. Hence $RS - T^2 \leq -8$, and the matrix is invertible for every average v .

The same argument can be written as a scalar elimination identity, with no division by the determinant:

$$\boxed{RE_{21} - 2TE_{17} = 2a_4(RS - T^2)}. \tag{15}$$

For a pair with equal squared moduli, $E_{17} = E_{21} = 0$, so (15) forces $a_4 = 0$. Multiplying the same identity by $-a_4$ gives its quantitative form:

$$-a_4(RE_{21} - 2TE_{17}) = 2a_4^2(R(-S) + T^2) \geq 16a_4^2.$$

This inequality applies to either sign of a_4 . With $a_4 = 0$, the first equation in (14) gives $a_6 = 0$, and (12) makes every remaining coefficient of h zero.

Both rows of the two-by-two system are coefficients of the same polynomial F , for the same average v and difference h . Thus their joint use applies to every candidate pair in (4). The remaining coefficients of F need not be solved: every actual solution must satisfy the equations already used, and those equations force $h = 0$.

5 Conclusion and the equivalent matrix statement

All eight possible first signal indices have been covered. In every case $h = 0$, so after normalization $f = g$. Undoing that normalization gives precisely a common unit phase. Section 2 then proves the finite-measurement theorem.

For completeness, the conclusion can also be expressed entirely in terms of Hermitian matrices. Define the real-linear map $\mathcal{A}_Q(H) = (q_j^* H q_j)_{j=1}^{26}$ on the Hermitian 8-by-8 matrices. Equality of the measurements of x and y is exactly the equation $\mathcal{A}_Q(xx^* - yy^*) = 0$.

The columns of Q span $\mathbb{C}^8$: a vector perpendicular to them would give a polynomial in U vanishing at 26 distinct points, hence the zero polynomial since its degree is at most 14. If a positive semidefinite Hermitian matrix H belongs to the kernel of $\mathcal{A}_Q$, each $q_j^* H q_j = 0$ implies $H q_j = 0$. Spanning then gives $H = 0$. The same applies after changing the sign of a negative semidefinite matrix. Finally, every indefinite Hermitian matrix of rank two can be written as $xx^* - yy^*$ with x, y not proportional, by its spectral decomposition. The theorem has excluded precisely these remaining possibilities. Therefore

$$\ker \mathcal{A}_Q \cap \{H = H^* : 0 < \text{rank } H \leq 2\} = \emptyset.$$

6 Scope and verification

The proof uses only the displayed polynomials, coefficient conjugation, elementary real inequalities, the normalization by the first nonzero coefficient, and Rolle's theorem. It does not depend on any prior unpublished classification or obstruction claim. The construction was motivated by choosing t^{14} , a polynomial with a repeated root, as an element shared by U and $U^\#$, where $U^\# = \{f^\# : f \in U\}$. No theory of such spaces is needed here.

The result concerns the specified frames and gives $m_{\mathbb{C}}(8) \leq 26$. It asserts neither optimality nor success for every or generic 26-vector frame. Numerical conditioning and a practical reconstruction algorithm with noise guarantees have not been established.

The supplementary archive provides the exact Gaussian-integer matrix `PR26_FRAME_QI.json`, the paper's sources, both verification implementations, and their logs. The original standard-library implementation uses exact rational arithmetic. An isolated rerun passed all 336 polynomial checks, seven matrix checks, and two negative controls, with mathematical results matching the supplied logs. These checks include the three missing coefficients, all substitutions and case identities, the actual frame entries, and rank calculations modulo a prime.

A separate SymPy implementation transcribes (1) without importing code from the original verifier. Its 633 finite checks passed: all 64 complex products, all 28 possible first-index pairs, the substitutions by fixed nonzero coefficients, the strict-sign identities, and all 208 matrix entries. The newly displayed symmetric two-by-two formulation is just a rearrangement of (14); its determinant agrees with (15). A separately supplied Wolfram log reports agreement for the three missing coefficients and eight principal coefficient identities and their combinations; that engine was not rerun during the local review.

The sampling argument, normalization, and case coverage have received a local logical review. Computational check counts do not count independent mathematical proofs. This is a proof candidate for critical examination: external independent expert review and proof-assistant verification have not been completed.

The mathematics, exploration, and verification were developed with substantial AI assistance. Detailed research records are retained separately. The complete proposed proof is given here, and the exact input and reproducibility scripts accompany it.